

Docker Lab-KVM/QEMU lab documentation

SETUP:-

- 1.Host- Linuxmint
- 2.Attacker-Linuxmint
- 3.Targets-Docker containers(DVWA,OWASP Juice shop,etc.),windows on kvm.

1.Host-

I dual booted windows 11 with mint as windows was taking too much memory and was becoming laggy and unusable, so I booted mint as I have limited hardware and I have used mint before on vm, mint was lightweight , user friendly , and most importantly stable to be used as daily OS, and it as it also uses apt packet manager , that meant I can use or get any tool that was on kali on mint, docker is native to linux and KVM too is more resource effecient than Virtualbox or Vmware as it's an bare metal virtual machine.

2.The Attacker

I intially planned to have kali linux on QEMU as an attacker and docker containers will have targets. But there was an input bug/misconfiguration that I spent one week trying to solve but it didn't work so instead I made the host OS(Linuxmint) as the attacker as well.

I set upped all the basic tools you need for pentesting like nmap, burpsuite, netcat, etc.

Turns out it was an better choice as i can now have windows as my target when i need on KVM, even though kali or parrot comes with pre installed tools and environment made for pentesting with limited hardware and resources it was more effecient setup.

3.Targets

I chose Docker containers for targets , as containers are really lightweight and I can very quickly destroy or create another one, currently I have only emulated and tested web application vulnerabilities.

I know docker containers can only emulate and test certain specific vulnerabilities , like web application, software dependency cve, misconfigured databases etc. It can't and should't be used to emulate kernel level exploits as it shares the same kernel as host OS, OS vulnerabilities etc.

So for that reason I chose KVM to emulate VMs to bridge the gap.

I will also be uploading the reports of the containers, CVEs I test in same repo.